3rd Milestone

Main idea

- Explain OrcaLink to people who may not know about whales or databases
- Focus on connecting family information and sighting records
- Make it clear that different records support different conclusions

Intro

- Start with the whales and the information we are studying
- Explain the goal before introducing technical details
- Define “matriline” as a maternal family group

Showing the existing sources

- Show what each source contributes
- CWR provides individual identity and family information
- Dryad links whale records to sightings
- The Whale Museum provides group sighting reports
- This comparison explains why combining the records needs care

Explaining the gap

- Some reports identify individual whales; others identify only groups
- Reporting a pod does not confirm every individual member
- Some whale identifiers come from group reports
- We need to preserve these differences when connecting records
- A shared sighting does not prove a close social relationship

Proposed model

- Use a simple diagram
- Show Whale and Matriline to explain family membership
- Show Kinship Assertion separately from Association
- Separate Individual Observation from Group Observation
- Connect observations to Sightings
- Include Data Source so the evidence behind records remains traceable
- Leave detailed attributes and participation limits out of the presentation sketch

Why it matters

- Researchers could examine family relationships alongside repeated sightings
- They could check which records support a connection
- Keeping identification information helps them interpret results appropriately
- Describe these as intended benefits because the model has not yet been evaluated

Knowledge graph

- Describe it as connected records with labeled relationships
- Explain the questions we aim to answer:
    - Family connections
    - Repeated co-sightings
    - Indirect connections

Outro

- This is a proposed model
- Describe potential associations as candidates for further investigation
- End with the goal
    - Making records easier to connect while keeping their meaning and evidence clear

- Intro - Bowen
- Explain - Aung

Our project is called OrcaLink. We're looking at Southern Resident killer whales and how to connect information about their families with records of where they've been seen.

The question we want to explore is: which whales have been recorded together, and how might those connections relate to their family relationships?

This could help researchers look across different records instead of checking each source separately. They could also see what evidence supports a connection.

In our literature review, we compared three sources. The Center for Whale Research records individual whale identities and family groups. The Dryad dataset connects whale records to sightings. The Whale Museum provides reports of groups seen at particular times and places.

These sources describe the same population, but at different levels of detail. That creates a problem when we combine them.

For example, a report might say that a pod was seen, without confirming every individual whale. If we treat that as individual identification, we could make the evidence look stronger than it actually is. Also, seeing two whales together does not automatically mean they have a close social bond.

Our proposed model keeps those differences clear.

In the diagram, kinship assertions describe family relationships between whales. We keep these separate from associations based on sightings or later analysis.

For sightings, we separate individual observations from group observations. This allows us to keep a useful group report even when individual identities are unknown.

We also keep the sources supporting the records. This helps us distinguish what was reported from connections we derive through analysis.

Next, we plan to turn this model into a relational database and then a knowledge graph, where whales and records become connected nodes with labeled relationships.

We aim to use it to explore family connections, repeated sightings together, and indirect connections between whales.

At this stage, we're proposing the model. Our goal is to connect the information while keeping clear what each record actually tells us.